generating function for Y is $m(t) = (pe^t + q)^n$, where $q = 1 - p$.

$$m(t) = E\left(\boxed{}\ \boxed{}\right) = \sum_{y=0}^{n} \binom{n}{y} \left(\boxed{}\ \boxed{}\right) q^{n-y} = (pe^t + q)^n$$

---

2. **[– / 7 Points]** DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 3.9.153.

Find the distributions of the random variables that have each of the following moment generating functions.

(a) $m(t) = \left[\left(\frac{1}{7}\right)e^t + \left(\frac{6}{7}\right)\right]^3$

The distribution is ---Select--- , with n = ____ and p = ____ .

(b) $m(t) = \dfrac{e^t}{2 - e^t}$

The distribution is ---Select--- , with p = ____ .

(c) $m(t) = e^{2(e^t - 1)}$

The distribution is ---Select--- , with λ = ____ .

---

3. **[– / 5 Points]** DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 3.9.155.

Let $m(t) = \left(\frac{1}{9}\right)e^{t} + \left(\frac{6}{9}\right)e^{2t} + \left(\frac{2}{9}\right)e^{3t}$.

(a) Find $E(Y)$.

$E(Y) =$ ______

(b) Find $V(Y)$.

$V(Y) =$ ______

(c) Find the distribution of Y.

Y	**1**	**2**	**3**
P(Y)			

4. **[– / 6 Points]** DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.2.013.

A supplier of kerosene has a 150-gallon tank that is filled at the beginning of each week. His weekly demand shows a relative frequency behavior that increases steadily up to 100 gallons and then levels off between 100 and 150 gallons. If Y denotes weekly demand in hundreds of gallons, the relative frequency of demand can be modeled by

$$f(y) = \begin{cases} y, & 0 \le y \le 1, \\ 1, & 1 < y \le 1.5, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find $F(y)$.

$$F(y) = \begin{cases} \_\_\_\_\_\_, & y \le 0, \\ \_\_\_\_\_\_, & 0 \le y \le 1, \\ \_\_\_\_\_\_, & 1 < y \le 1.5, \\ \_\_\_\_\_\_ \end{cases}$$

$y > 1.5$

,

(b) Find $P(0 \le Y \le 0.4)$.

(c) Find $P(0.4 \le Y \le 1.4)$.

5. [– / 10 Points] DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.2.017.

The length of time required by students to complete a one-hour exam is a random variable with a density function given by

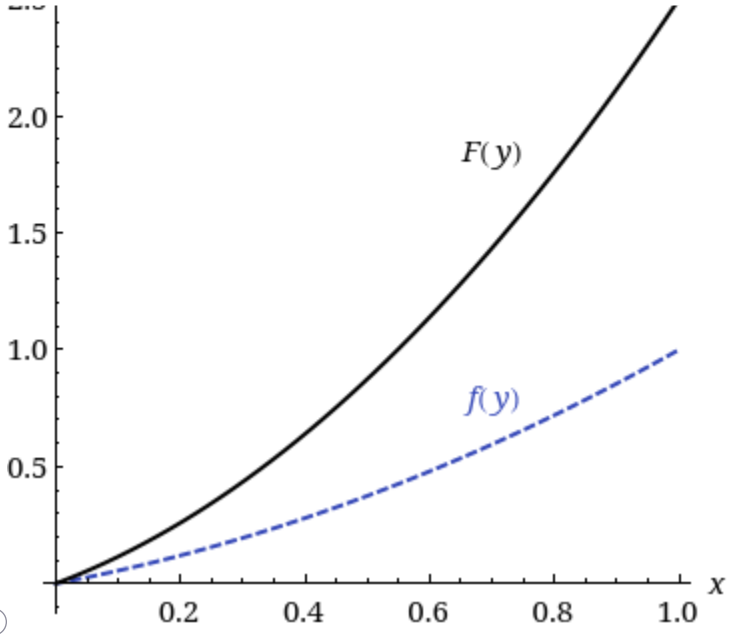
$$f(y) = \begin{cases} cy^2 + y, & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

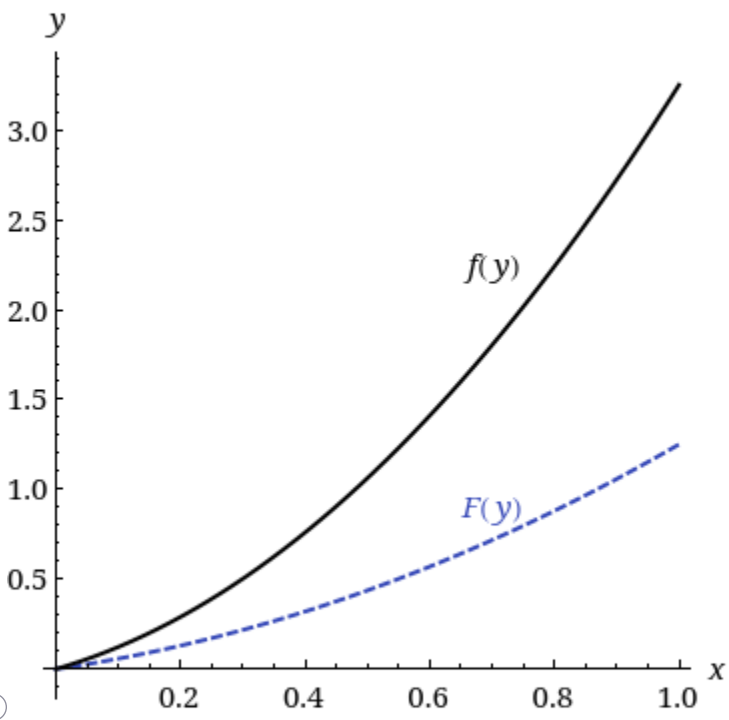
(a) Find c.

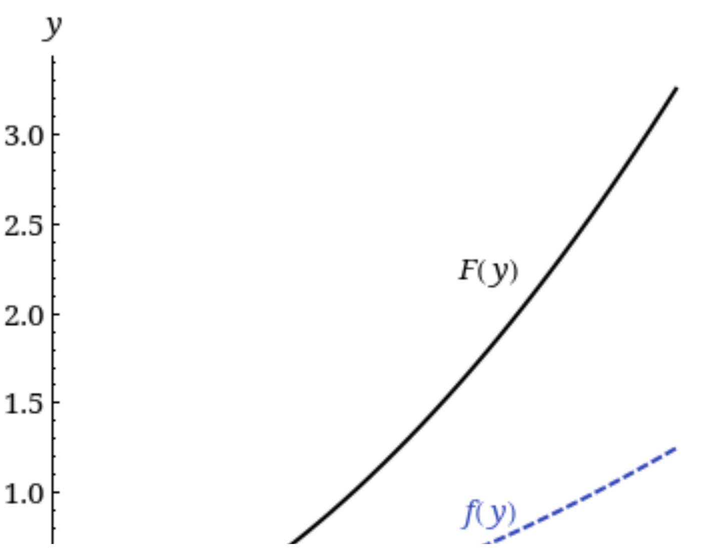
(b) Find $F(y)$.

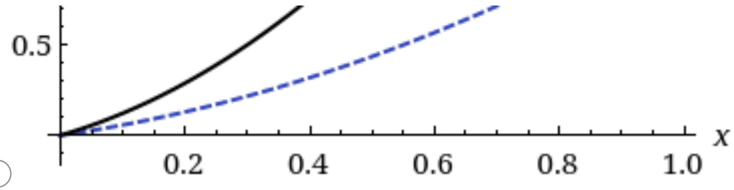
$$F(y) = \begin{cases} \text{______}, & y < 0 \\ \text{______}, & 0 \le y \le 1, \\ \text{______}, & y > 1 \end{cases}$$

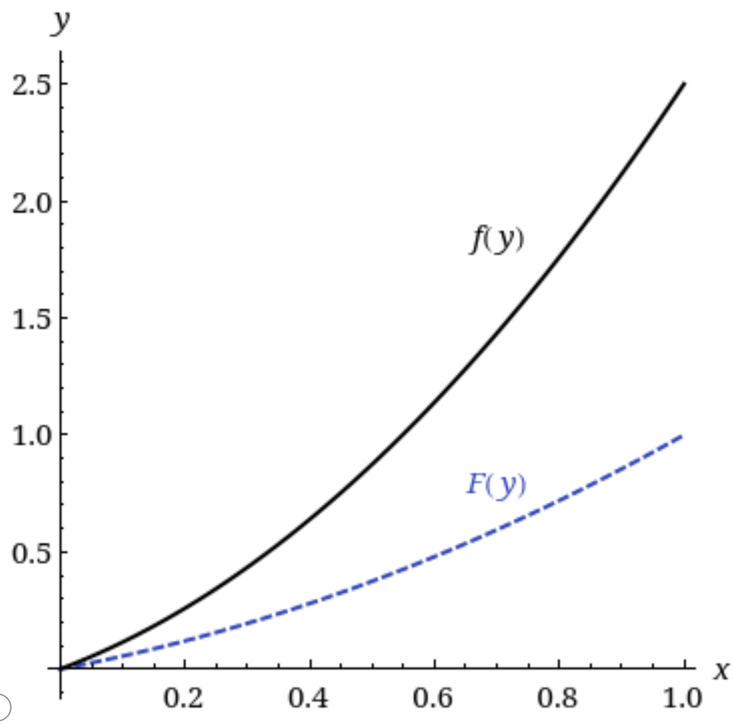
(c) Graph $f(y)$ and $F(y)$.


2.0
1.5
1.0
0.5
F(y)
f(y)
x
0.2
0.4
0.6
0.8
1.0


y
3.0
2.5
2.0
1.5
1.0
0.5
f(y)
F(y)
x
0.2
0.4
0.6
0.8
1.0


y
3.0
2.5
2.0
1.5
1.0
F(y)
f(y)





(d) Use $F(y)$ in part (b) to find $F(-1)$, $F(0)$, and $F(1)$.

$F(-1)$ =

$F(0)$ =

$F(1)$ =

(e) Find the probability that a randomly selected student will finish in less than half an hour. (Round your answer to four decimal places.)

(f) Given that a particular student needs at least 15 minutes to complete the exam, find the probability that she will require at least 30 minutes to finish. (Round your answer to four decimal places.)

6. **[– / 2 Points]** DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.3.021.

Y has a density function

$$f(y) = \begin{cases} \frac{7}{2}y^6 + y, & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of Y. (Round your answers to four decimal places.)

$E(Y)$ = [ ]

$V(Y)$ = [ ]

---

7. [– / 2 Points] DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.3.027.

For certain ore samples, the proportion Y of impurities per sample is a random variable with density function

$$f(y) = \begin{cases} \frac{7}{2}y^6 + y, & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

The dollar value of each sample is $W = 4 - 0.4Y$. Find the mean and variance of W. (Round your answers to four decimal places.)

$E(W)$ = [ ]

$V(W)$ = [ ]

---

8. [– / 1 Points] DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.3.031.

Daily total solar radiation for a specified location in Florida in October has a probability density function given by

$$f(y) = \begin{cases} \left(\frac{3}{4}\right)(y - 3)(5 - y), & 3 \le y \le 5, \\ 0, & \text{elsewhere,} \end{cases}$$

with measurements in hundreds of calories. Find the expected daily solar radiation for October, in hundreds of calories.

$E(Y)$ = [ ] hundred calories

---

9. [– / 2 Points] DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.4.047.

The failure of a circuit board interrupts work that utilizes a computing system, until a new board is delivered. The delivery time, Y, is uniformly distributed on the interval two to seven days. The cost of a board failure and interruption includes the fixed cost c_0 of a new board and a cost that increases proportionally to Y^2. If C is the cost incurred, $C = c_0 + c_1Y^2$.

(a) Find the probability that the delivery time exceeds four days.

(b) In terms of c_0 and c_1, find the expected cost associated with a single failed circuit board.

10. **[– / 1 Points]** DETAILS MY NOTES ASK YOUR TEACHER

WackerlyStat7 4.4.049.

A telephone call arrived at a switchboard at random within a one-minute interval. The switch board was fully busy for 5 seconds into this one-minute period. What is the probability that the call arrived when the switchboard was not fully busy? (Round your answer to four decimal places.)

Home My Assignments Request Extension